

ARCHITECTURE OVERVIEW

The Inaya Ecosystem, End to End

How the storage protocol, the applications built on it, and the AI layer all fit together — the shape of the system, without the code.

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SECTION 01

The Shape Of The System

Inaya is one ecosystem built on two connected layers. The bottom layer is a real DePIN protocol — people encrypt their own files, independent operators store and verify them, and a blockchain anchors who owns what. The top layer is a set of applications — web, mobile, two desktop apps, and a business SaaS product — that most users interact with without ever knowing a blockchain is involved.

THE PROTOCOL LAYER

Smart contracts that handle ownership, staking, node payouts, and threat verification. This is where the token economics live, and where independent node operators actually earn.

THE APPLICATION LAYER

Everything a person actually opens: the storage dApp, the Business Workspace for companies (including its own operations, finance, and HR tooling), the mobile app, two desktop apps, a public security page, an investor data room, and four AI assistants — all talking to the same backend.

SECTION 02

Storage — How A File Actually Gets Protected

This is the core promise: nobody but the file's owner can ever read it — not Inaya, not the node storing it, nobody.

1. **Encrypt on your device.** Before a file ever leaves the user's browser or phone, it's encrypted using a key derived from their own passkey. That passkey is never sent anywhere.
2. **Split it in two.** The encrypted file is split into two halves. Neither half means anything on its own — you need both to reconstruct anything.
3. **Store both halves.** The two halves get pinned to decentralized storage (IPFS/Pinata), independent of each other.
4. **Anchor ownership on-chain.** A record — who owns this file, where its two halves live, a fingerprint proving it hasn't been tampered with — gets written to the blockchain. This is the permanent, tamper-evident receipt.
5. **Reverse it to read.** Downloading does the same thing backwards: look up the on-chain record, fetch both halves, put them back together, decrypt with the owner's passkey. Nothing about this process gives Inaya, a node operator, or anyone else a way to see the plaintext.

You don't have to take Inaya's word for any of this. The encryption code above is published as open source (@inaya-network/custody-sdk on npm) and used identically by the web app and the mobile app — not a separate, unverifiable implementation behind the scenes. Every release publishes a cryptographic hash anyone can independently reproduce and check, plus a second copy pinned to IPFS so the code itself, not just Inaya's word, identifies what you're running. Full verification steps are public.

SECTION 03

Node Operators — Who Actually Stores The Data

Independent people and organizations run node software that registers them on-chain and reports their capacity. In exchange, they earn USDT and \$INAYA — for storage service, and separately, for contributing to a decentralized threat-intelligence network (see Section 05).

- Registration and identity live on-chain — a node's stake, tier, and track record are public and verifiable.
- Payouts are deliberately delayed and verifier-attested rather than instant and self-reported — this protects the system from a single bad actor (or a single compromised key) draining funds in one move.
- Today, the software an operator runs handles registration and reporting — it doesn't yet handle the actual physical hosting of file shards. That's a known, stated gap in the current build, not a hidden one.

SECTION 04

The Web dApp — Where Consumers And The Protocol Meet

The main website is where an individual user does everything token-and-storage related: gets testnet tokens from a faucet, uploads/downloads files in the Sovereign Vault, stakes \$INAYA for yield, tracks their activity on a dashboard, refers friends (email + identity verification, no wallet needed for that part), and — for larger customers — buys an annual Corporate Reserve storage plan, paid in USDT and settled on-chain.

Staking has real lock-period tiers (flexible, 30-day, 90-day) with better rewards for longer commitments — this is enforced on-chain, not just a UI label.

SECTION 05

Business Workspace — A Second Product On The Same Foundation

Not every customer wants a blockchain product. Business Workspace is a genuine B2B SaaS tool — organizations, departments, projects, documents — with real approval workflows, granular permissions, secure sharing links, and a permission-aware AI assistant. Sign-in is just an email or a Google account. No wallet, no tokens, no blockchain vocabulary anywhere in the experience.

It has grown well beyond documents: real business operations (Tasks, CRM, Procurement, Inventory) and a Finance & HR layer (invoicing, expense approval, employee records, leave management) now run on the exact same organization/department/permission foundation described above — every new module additive, nothing about the base system had to be restructured to add them. Finance & HR is explicitly a testnet demonstration layer today, not regulated banking or payroll.

Why this matters strategically. It's a second, independent revenue engine — priced and sold like any other business software, evaluated by a buyer who will never ask what chain it runs on — while still running on top of the same infrastructure and the same encryption guarantees as the consumer storage product.

SECTION 06

Security Layer — A Decentralized Threat Network

Marketed publicly as "Inaya Firewall." The same trust model as the storage layer, applied to threat intelligence instead of files: independent nodes report suspicious websites, the network only trusts a threat once enough independent, reputable reporters agree, and confirmed verdicts get permanently recorded on-chain so they can't quietly be changed later.

- A public website page lets anyone — no account needed — check a domain, see network stats, and ask an AI assistant about how it all works.
- The mobile app can check destinations before you trust them and keep your own personal block/allow list.
- The desktop apps can, on Windows, actually block a confirmed-malicious address at the operating-system firewall level, not just inside the app window.

SECTION 07

The AI Layer — Four Assistants, One Pattern, Different Rules

All four assistants (Docs, Business, Security, Learn Tutor) are built the same technical way — they can call small, purpose-built tools to look up real data before answering. What's different is the philosophy each one follows.

ASSISTANT	ITS ONE RULE
Docs Assistant	Answer grounded in the project's own real documentation — retrieves the relevant passage first, then answers from it, and cites where the answer came from.
Business Assistant	Never show a user data they aren't actually permitted to see — every answer respects real document/project/finance/HR permissions.
Security Assistant	Never invent a threat verdict — every specific claim must be backed by real, verified network data.
Learn AI Tutor	The opposite of the other two — teach freely using its own knowledge, like a real tutor would; only use tools to check the user's own saved videos and progress.

How the Docs, Security, and Learn assistants find the right passage. A shared retrieval layer (RAG) indexes the project's real documentation and FAQs, embeds it for semantic search, and combines that with plain keyword search — so "how does the pricing work" and "pricing structure" both find the same real answer, not a static block of hardcoded text that goes stale the moment the product changes.

SECTION 08

Mobile & Desktop — Same Backend, Different Doors

None of the client apps are separate products with their own logic — they're different doors into the exact same backend and the exact same data.

- The mobile app is a full superset — everything the web dApp does, plus the Business Workspace, Learn, and Security, in one app.
- The two desktop apps are deliberately thin — each one is just the real website, running inside a lightweight native window with a system tray icon, native notifications, and auto-updates layered on top. There is no separate desktop-only feature set to keep in sync.

SECTION 09

Oracle & Automation — Infrastructure That Runs Itself

A third independent subsystem: an on-chain registry of approved data sources, and a keeper that executes pre-approved contract actions automatically — under smart-contract rules, never arbitrary admin commands. Live on BSC Testnet today, publicly verifiable at inayanetwork.com/automation.

- The live demo isn't simulated: the price feed reads the real PancakeSwap testnet pool for INAYA/USDT, and the automated action is a real, already-existing node-settlement release function on InayaNodeRegistry — the first run found and released an actual previously-unclaimed payout.
- If the data feeding an automated action goes stale, the system skips that action rather than acting on unverified information — proven live, not just claimed.
- The keeper never holds broad authority — it can only submit data it's explicitly registered for, and only call functions that are already safe for anyone to call.

SECTION 10

How It All Actually Connects

One backend serves every surface — the website, the mobile app, and both desktop apps all call the exact same API. There's no duplicated logic to keep in sync between platforms; a fix or feature on the backend is instantly live everywhere.

The protocol (contracts, node operators, encryption) and the applications (dApp, Business Workspace, mobile, desktop, AI) are cleanly separated — most of what a user touches day-to-day never has to think about the blockchain underneath it, even though it's doing real work a layer down.

Everything currently runs on BNB Chain Testnet. Nothing described in this document is live on mainnet yet — that's the next major milestone, not a past one.